

SHREE NIPANE

Vellore, India | shreenipane@gmail.com | shreenipane.github.io/portfolio | linkedin.com/in/shreenipane | github.com/shreenipane

EDUCATION

Vellore Institute of Technology (VIT), Vellore 2024 – 2028
Bachelor of Technology in Computer Science and Engineering (expected graduation: 2028)
Class XII: Swarajya Senior Secondary School, Alwar | 82.5% | 2024
Class X: The Lexicon International School, Wagholi, Pune | 72.5% | 2022

KERNEL AND SYSTEMS PROJECTS

Verdict | In-kernel network packet classifier for Linux | in progress, code private
C99, eBPF, Linux kernel, GCC, Make, QEMU/KVM, Linux 7.1

- Built a classifier that decides whether a network packet is malicious inside the Linux kernel, at the earliest point on the receive path. It loads and runs on a real kernel, not a simulator.
- Measured **250 ns** of CPU time per classified packet with BPF_PROG_TEST_RUN in throwaway QEMU/KVM guests running the host's kernel.
- Verified with differential and mutation tests: all **2,080,804** arithmetic cases checked exhaustively against a separate implementation with zero mismatches; **114,401** real network frames replayed with zero faults; **22 of 23** injected bugs caught (the survivor was equivalent code).
- Implementation details are withheld while a patent application is considered. Accuracy on labelled attack traffic is the next test.

VAGUS | cgroup v2 resource and QoS manager for Linux servers | github.com/shreenipane/VAGUS
Python, cgroups v2, PSI, /proc, systemd, eBPF (cgroup_skb), bubblewrap, pytest

- Built a controller that reads cgroup v2 and /proc telemetry (cpu.stat, memory.current, io.stat, PSI) and writes cpu.max, cpu.weight and memory.high inside delegated systemd subtrees to protect latency-critical services.
- Cut a web service's P99 latency from **258 ms to 9.0 ms** (8.4 ms uncontended) beside a CPU hog and UDP flood (6 arms x 4 reps), after withdrawing an earlier result whose load generator was starved.
- Made writes safe on a live host: path validation and traversal rejection, dry-run default, crash-consistent journal with exact revert, a single-writer lock and a PSI-gated rollback loop; **170+** tests run in a bubblewrap sandbox.
- Prototyped an eBPF cgroup_skb ingress program to charge NET_RX softirq time to the owning cgroup (built; live run pending root access). Telemetry costs **0.6%** of one core across 139 cgroups.

Also built: PulseGlass, on-device webcam heart-rate estimation in TypeScript and React with a tested DSP core (33 unit tests) | live demo: pulse-glass.vercel.app | github.com/shreenipane/pulse-glass

TECHNICAL SKILLS

Languages: C (primary), Python, C++, Java, SQL, TypeScript
Kernel and OS: eBPF and the BPF verifier, cgroups v2, PSI, softirq accounting, systemd, /proc and cgroupfs, operating system fundamentals
Networking: TCP/IP header parsing, packet processing, pcap/pcapng datasets
Tools: Git, GitHub, GCC, Make, QEMU/KVM, bubblewrap, pytest; daily Linux user (Fedora)
Testing and Practice: Differential, exhaustive and mutation testing; performance measurement; design docs and engineering logs
Also: Data structures and algorithms (~150 LeetCode problems), system design, PyTorch, agentic AI

CERTIFICATION

AWS Certified AI Practitioner (AIF-C01) | Amazon Web Services